

IGNITE64 Final Submission: Project Summary

Project Name: Biomedical Signal Visualizer ([BIOCORE AI Platform](#))

Team: is.medicine

Modern clinical monitoring and medical education face a dual crisis: they are highly fragmented, prohibitively expensive, and increasingly reliant on opaque “black-box” artificial intelligence systems. In critical care and diagnostics, the tendency of generative AI to hallucinate poses severe ethical, legal, and patient safety risks that modern hospitals cannot afford. Furthermore, access to applied physiological simulation and biomedical hardware integration remains restricted for students and researchers globally, creating a massive barrier to democratized healthcare education. We urgently needed a solution that democratizes medical intelligence while strictly adhering to data privacy, rigorous clinical safety, and emerging cyber law standards.

To address this, we developed the *Biomedical Signal Visualizer* (BIOCORE AI Platform), a system that transcends traditional telemetry to become a comprehensive, open-source computational physiology ecosystem. We built a unified environment where users can learn pathophysiology, analyze real biomedical signals, including ECG, EMG, EEG, and SpO₂, and seamlessly connect local biomedical hardware via serial ports. At its core, the platform features an advanced interactive “Digital Twin” that allows users to manipulate physiological parameters in real-time, instantly observing multi-systemic clinical effects. By fusing these metrics in our Proprietary Biomarkers Lab, we bridge the gap between raw biometric data and actionable, explainable clinical education.

To ensure absolute ethical AI compliance and data privacy—critical priorities in modern health-tech and cyber law—we engineered a robust, dual-layer intelligence architecture. First, a classical Machine Learning pipeline, trained locally on the open MIT-BIH Arrhythmia Database, provides highly reliable baseline signal anomaly detection. Second, and most importantly, we developed a Deterministic Clinical Decision Support System (CDSS). Instead of relying on generative Large Language Models for critical diagnoses, this local, rules-based engine evaluates over 50 intensive-care logic gates derived from established medical protocols, encompassing vitals, laboratories, and anthropometry. It outputs transparent, hallucination-free triage reports, guaranteeing that no Protected Health Information (PHI) is ever exposed to external cloud servers, thus ensuring total patient confidentiality.

Developed entirely in Python, the platform is 100% open-source, modular, and container-ready. Its flexible architecture scales effortlessly: it can run on a local medical student’s laptop processing live EMG data via an affordable ESP32 microcontroller, or it can be expanded into a fully deployed cloud infrastructure for institutional cohort research and remote patient monitoring. This highly scalable design fosters a truly open ecosystem, inviting continuous contribution, auditability, and peer review from the global digital health and cybersecurity communities.

AI Tool Disclosure (Rule 04 Compliance): In strict compliance with IGNITE64 Transparent AI guidelines, team *is.medicine* discloses the use of OpenAI ChatGPT and Google Gemini (latest versions) strictly as conversational assistants for logic ideation, rapid code debugging, and structuring physiological knowledge. GitHub and VS Code (with AI extensions) were used for version control and syntax optimization. To guarantee clinical safety, all core physiological logic, the deterministic engine architecture, and final system integration were directed, written, and validated manually by the human team. Generative AI was purposefully excluded from the clinical diagnostic pipeline.